

Geometry
Unit 10
Lesson 2 Practice

Name Answer Key
Date
Period

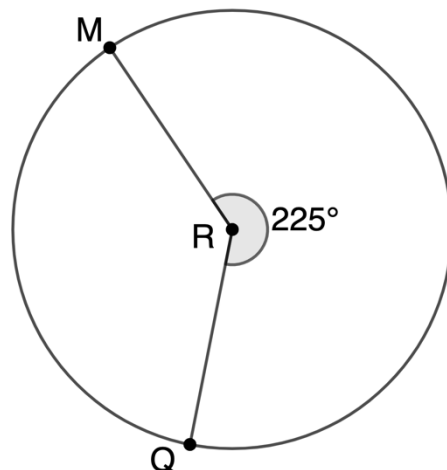
For questions 1-2, Circle R is shown where points M and Q are on the circle. The measure of central angle QRM is 225° .

1. What is the ratio of the central angle, $\angle QRM$, to the whole circle? Simplify the ratio.

$$\frac{225}{360} = \frac{5}{8}$$

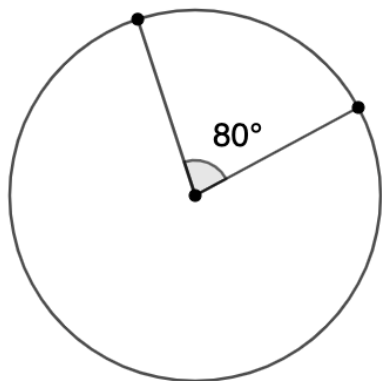
2. What fraction of the circle's circumference does $\widehat{MQ}$ represent?

$$\frac{225}{360} = \frac{5}{8} \text{ for the larger arc; } \frac{135}{360} = \frac{3}{8} \text{ for the smaller arc}$$



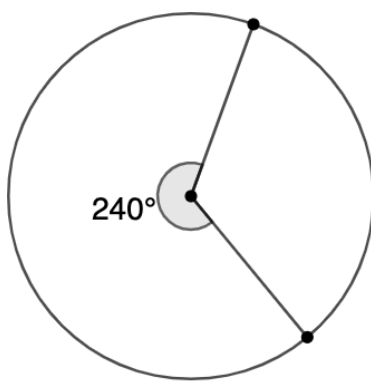
For questions 3-8, find the ratio of the arc length to the circumference of the circle.

3.



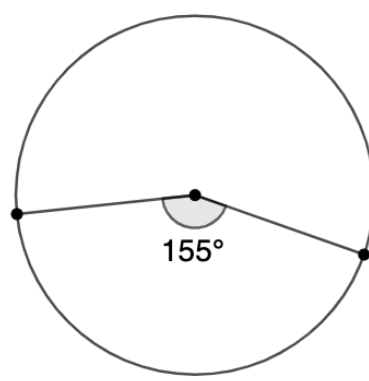
$$\frac{80}{360} = \frac{2}{9}$$

4.



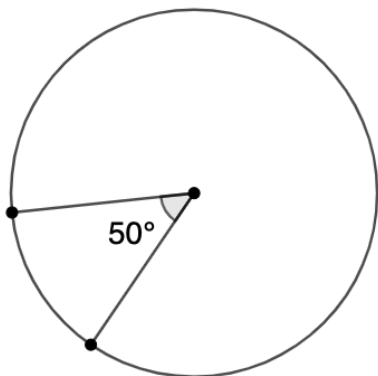
$$\frac{240}{360} = \frac{2}{3}$$

5.



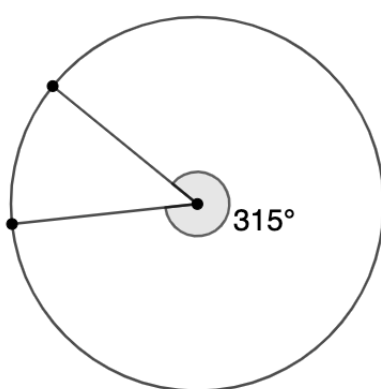
$$\frac{155}{360} = \frac{31}{72}$$

6.



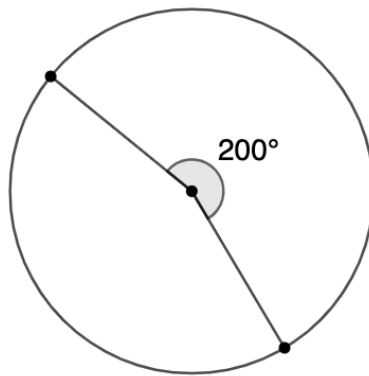
$$\frac{50}{360} = \frac{5}{36}$$

7.



$$\frac{315}{360} = \frac{7}{8}$$

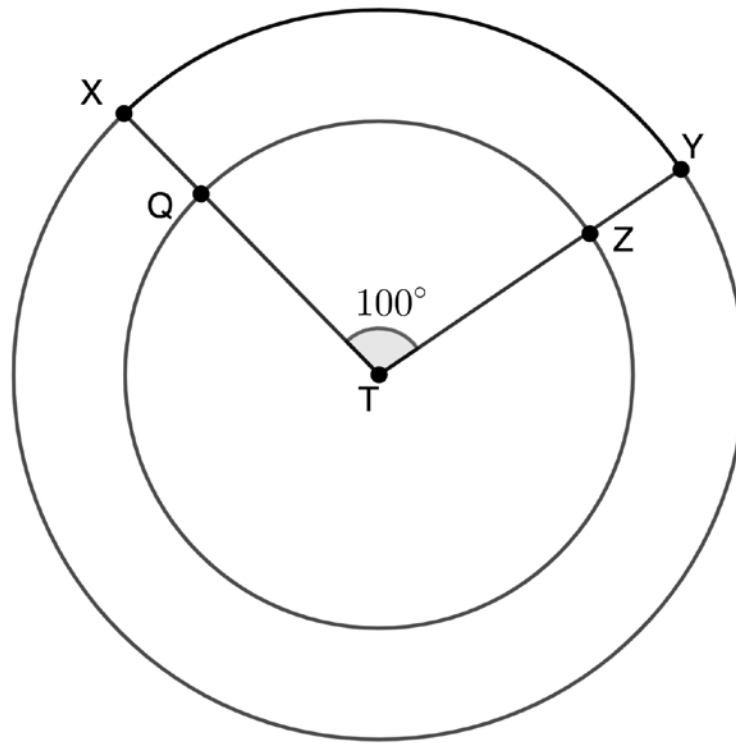
8.



$$\frac{200}{360} = \frac{5}{9}$$

For questions 9-10, consider the following information.

Points X and Y lie on Circle T . The radius $\overline{TX}$ measures 10 inches (in.). A smaller Circle T with a radius measure of 7 in. rests inside the larger Circle T , where points Q and Z lie on the smaller circle. The smaller Circle T has central angle $\angle QTZ$ that measures 100° and the larger Circle T has central angle $\angle XTY$ that measures 100° .



9. What is the ratio of the arc length to circumference of Circle T with radius measure of 10 in.?

$$\frac{100}{360} = \frac{5}{18}$$

10. What is the ratio of the arc length to circumference of Circle T with radius measure of 7 in.?

$$\frac{100}{360} = \frac{5}{18}$$